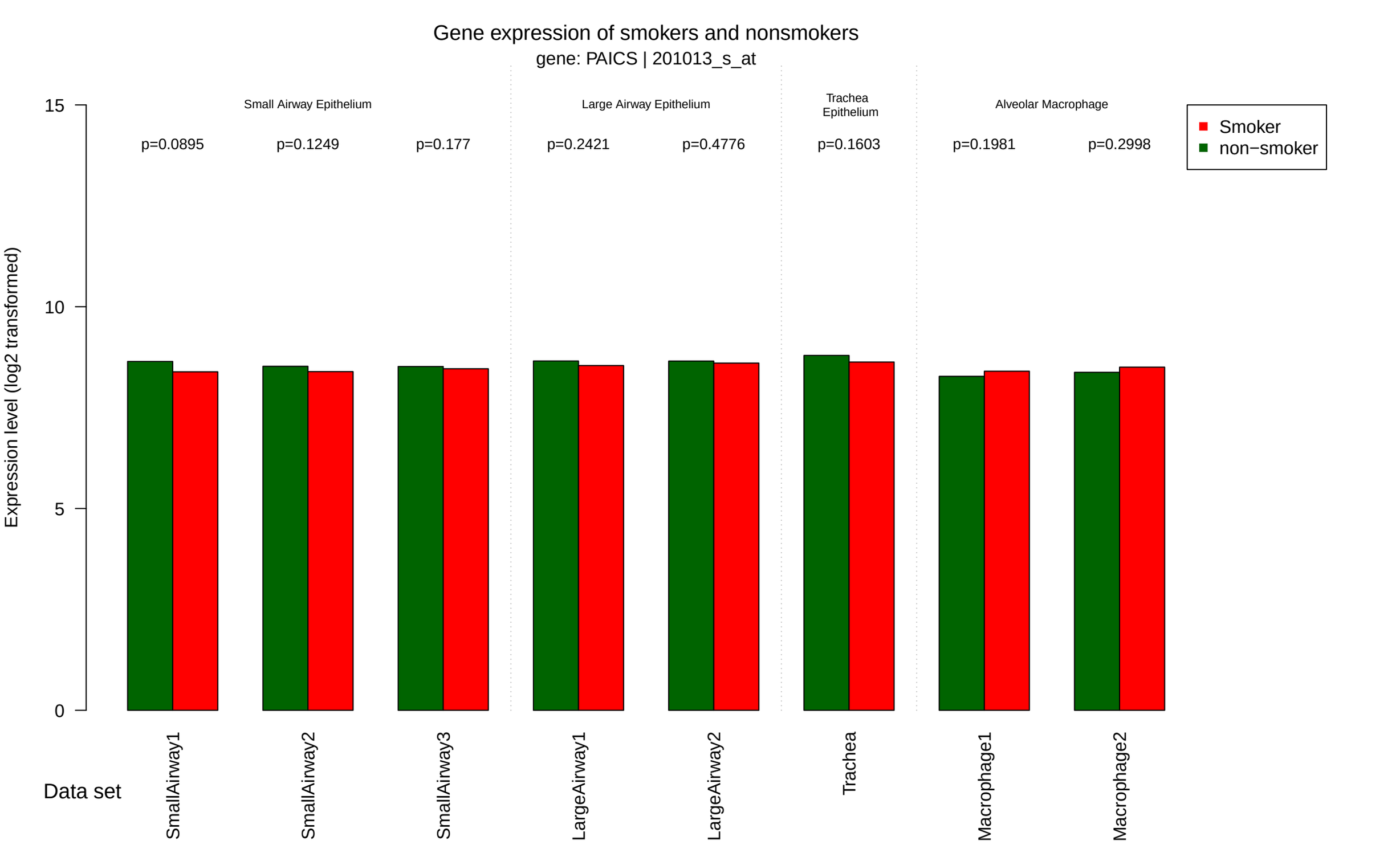
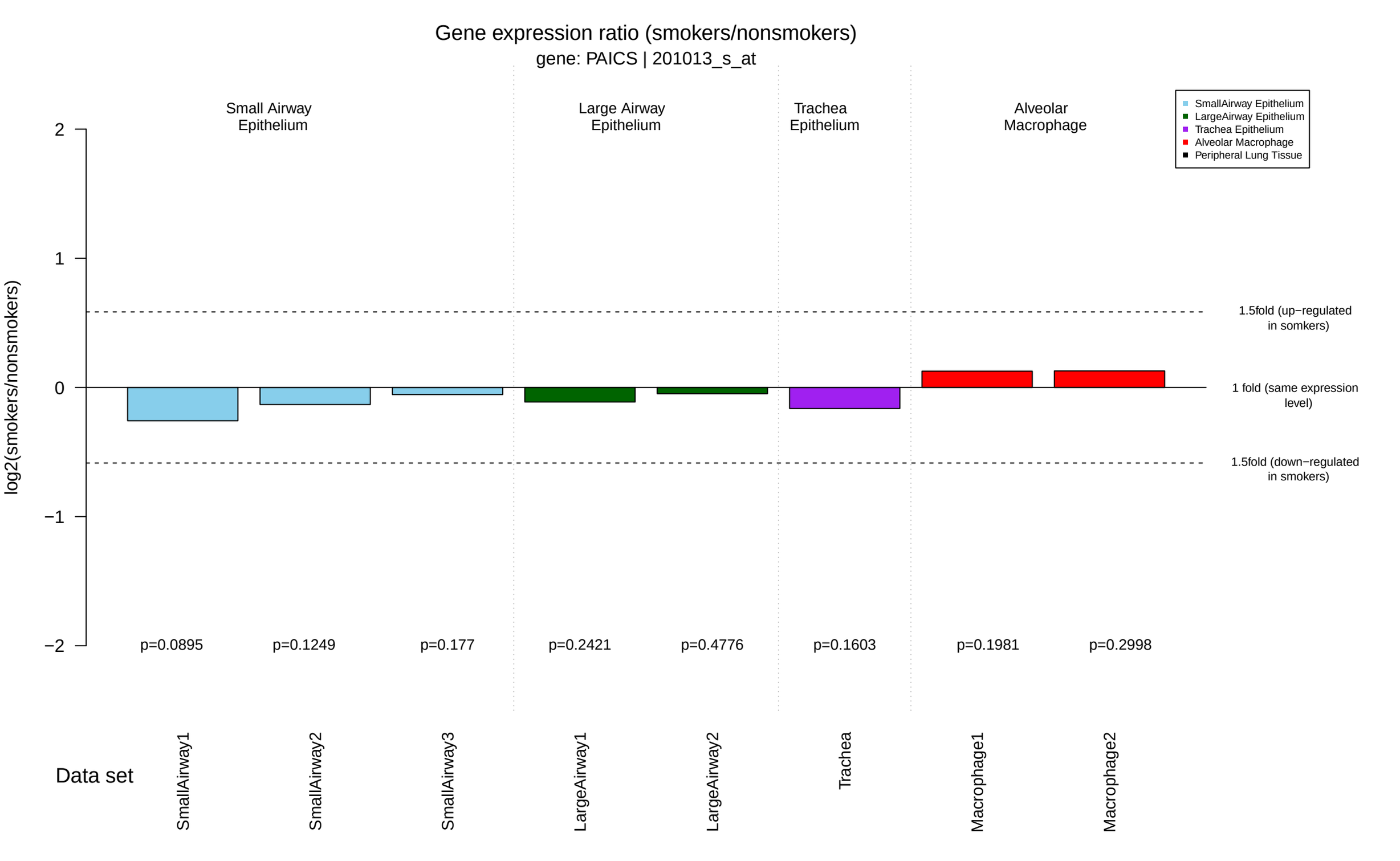
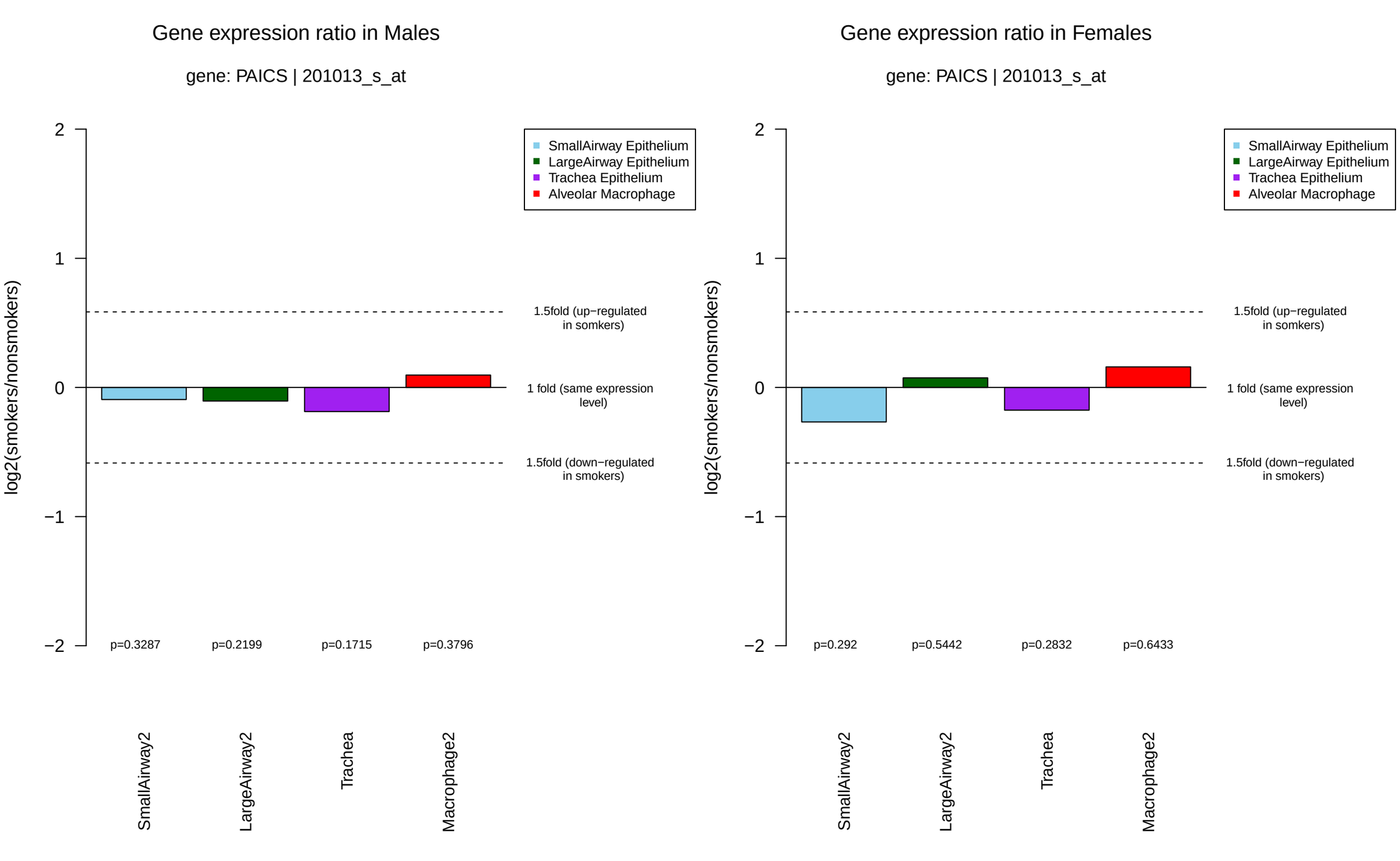
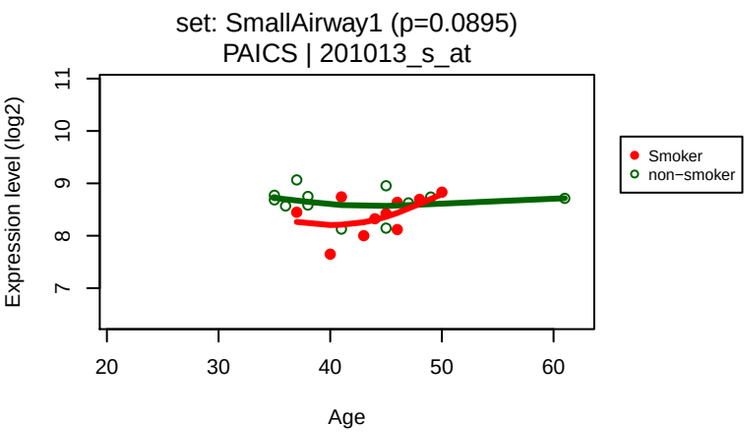
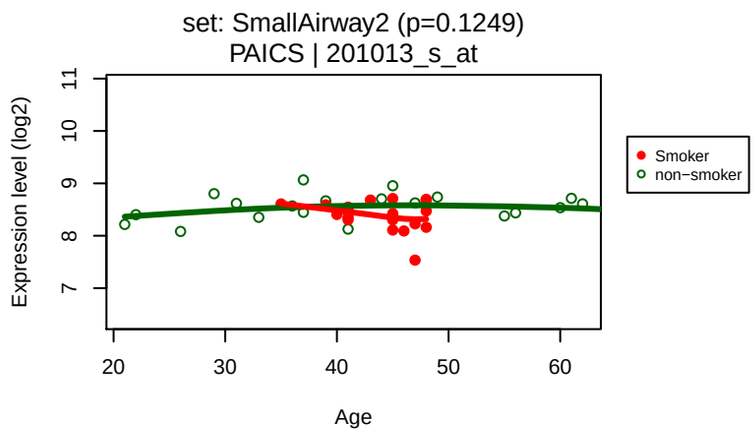
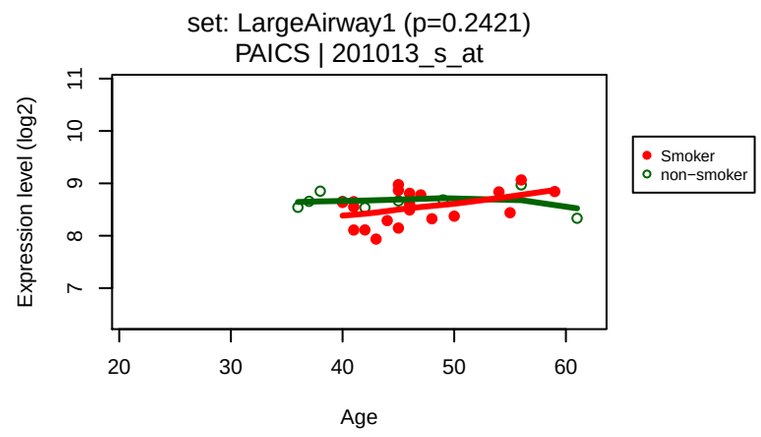

Gene expression of smokers and nonsmokers
gene: PAICS | 201013_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0895
p=0.1249
p=0.177
p=0.2421
p=0.4776
p=0.1603
p=0.1981
p=0.2998
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

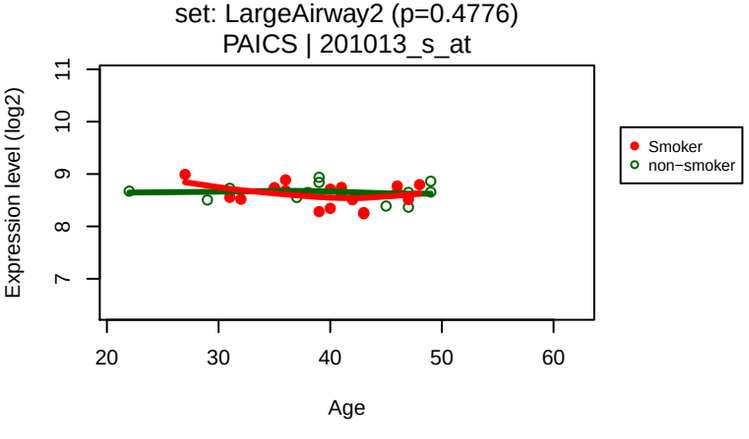

Gene expression ratio (smokers/nonsmokers)
gene: PAICS | 201013_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0895
p=0.1249
p=0.177
p=0.2421
p=0.4776
p=0.1603
p=0.1981
p=0.2998
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

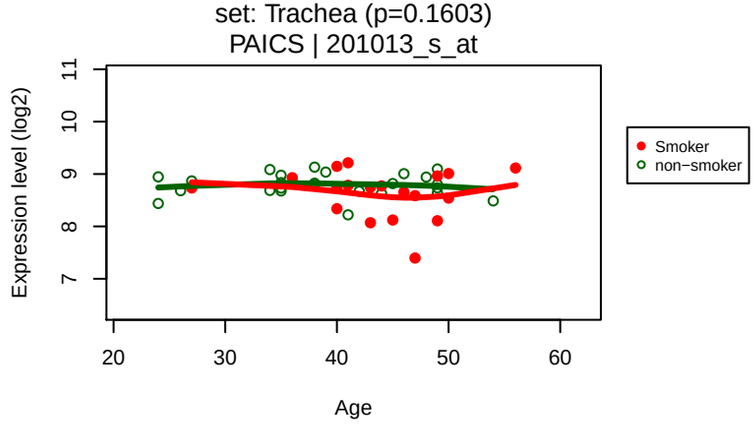

Gene expression ratio in Males
gene: PAICS | 201013_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3287
p=0.2199
p=0.1715
p=0.3796
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAICS | 201013_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.292
p=0.5442
p=0.2832
p=0.6433
SmallAirway2
LargeAirway2
Trachea
Macrophage2

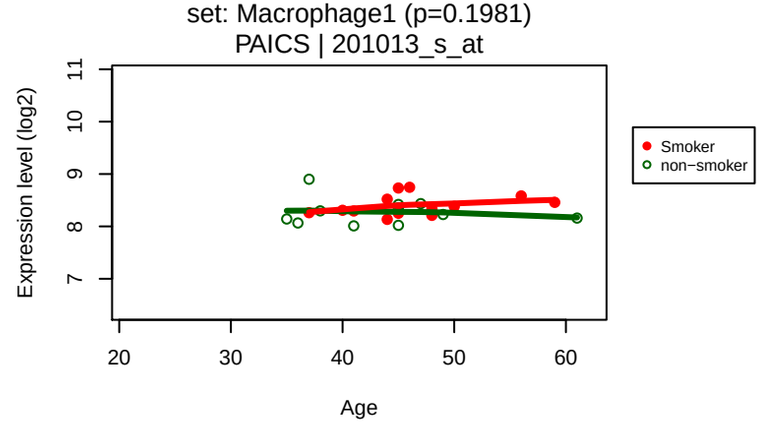

set: SmallAirway1 (p=0.0895)
PAICS | 201013_s_at
Expression level (log2)
Age
Smoker
non−smoker

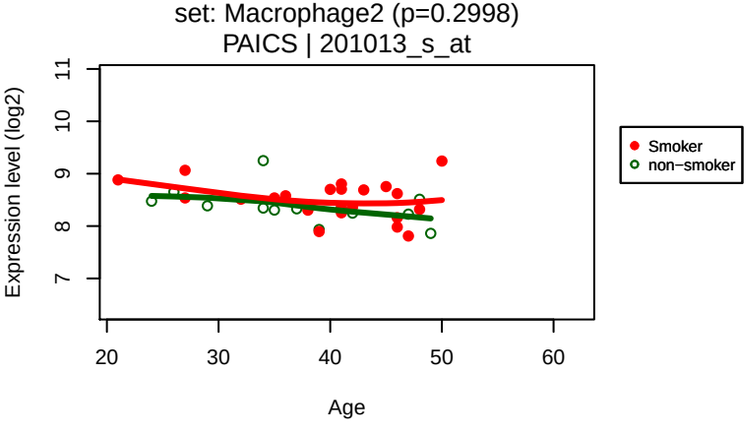

set: SmallAirway2 (p=0.1249)
PAICS | 201013_s_at
Expression level (log2)
Age
Smoker
non−smoker

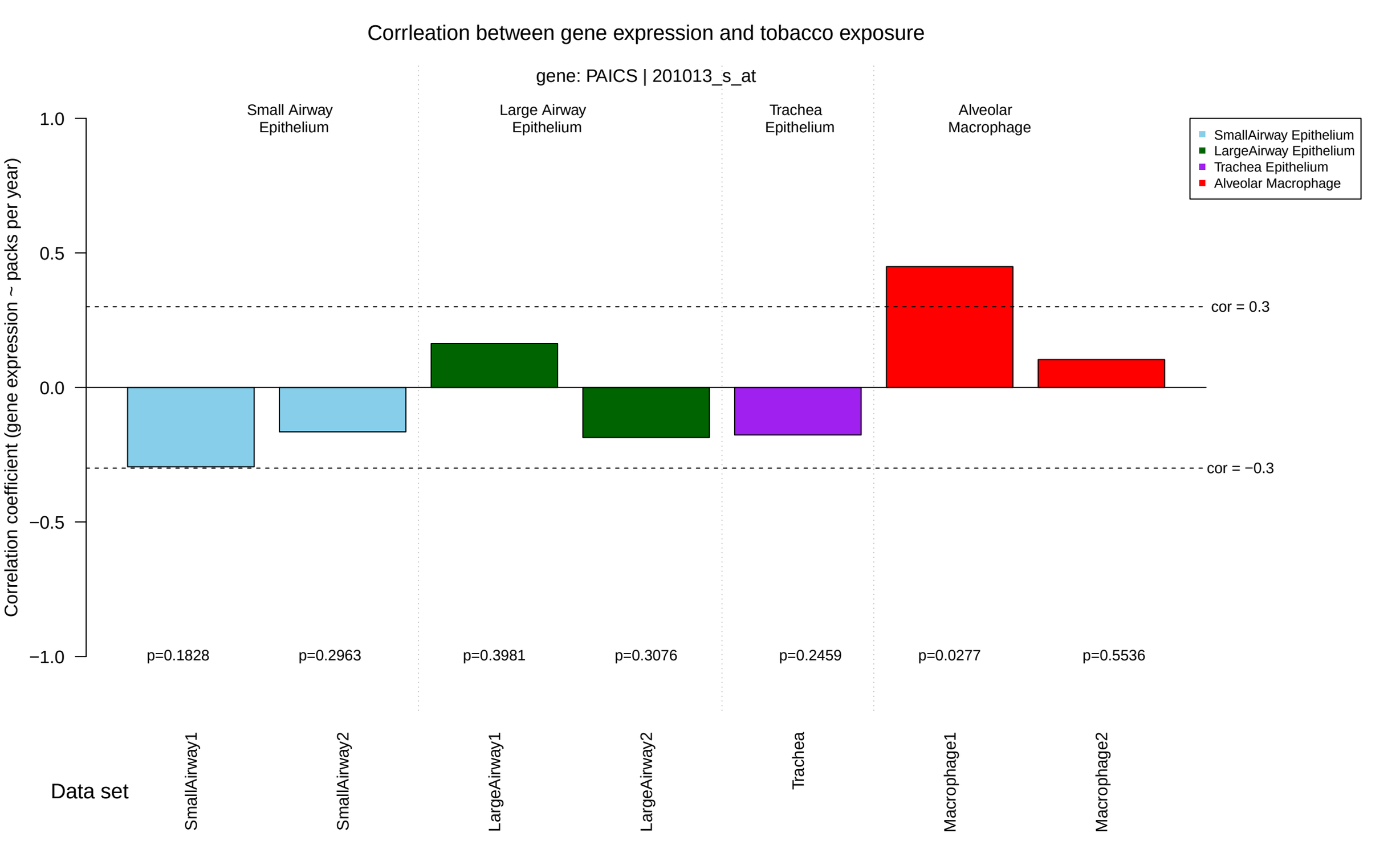

set: LargeAirway1 (p=0.2421)
PAICS | 201013_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4776)
PAICS | 201013_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1603)
PAICS | 201013_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1981)
PAICS | 201013_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2998)
PAICS | 201013_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAICS | 201013_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1828
p=0.2963
p=0.3981
p=0.3076
p=0.2459
p=0.0277
p=0.5536
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2